Tonmya_Prescriptions

2026-06-28

Some simple models based on Tonmya sales

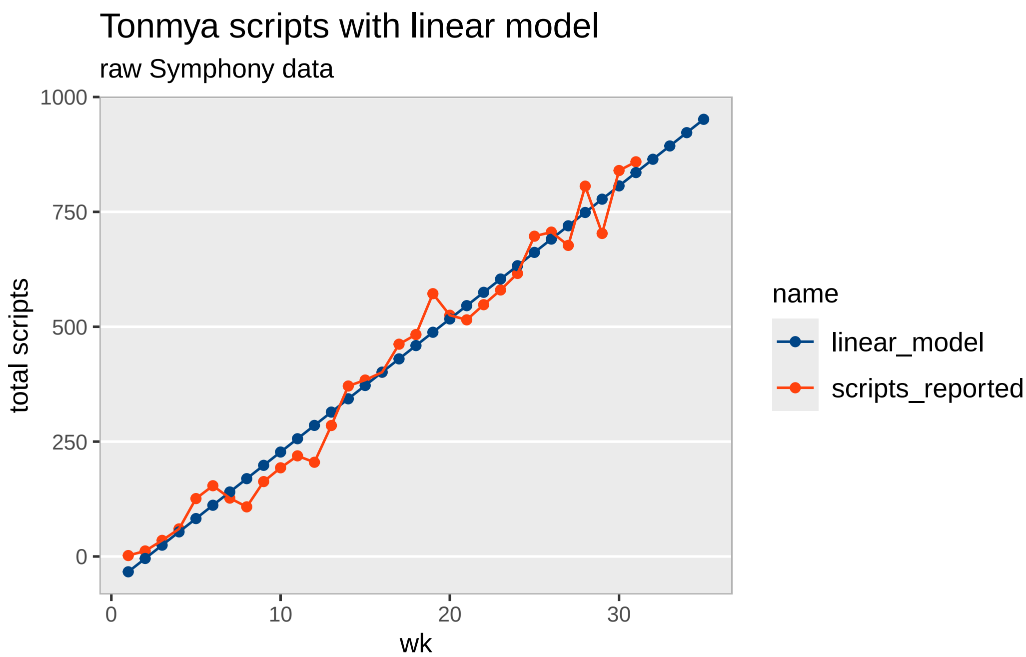
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

scripts Projection - at weekly rate - ***linear model***

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max 
-80.242 -30.060   8.048  29.885  84.012 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -62.3226    14.7251  -4.232 0.000212 ***
wk           28.9637     0.8033  36.055  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40 on 29 degrees of freedom
Multiple R-squared:  0.9782,    Adjusted R-squared:  0.9774 
F-statistic:  1300 on 1 and 29 DF,  p-value: < 2.2e-16
```

Scripts, Linear Model

wk	weekly_seq	total reported	linear predicted	annual
1	2025-11-14	2	-33.4	1,506
2	2025-11-21	12	-4.4	3,012
3	2025-11-28	35	24.6	4,518
4	2025-12-05	60	53.5	6,024
5	2025-12-12	126	82.5	7,531
6	2025-12-19	154	111.5	9,037
7	2025-12-26	127	140.4	10,543
8	2026-01-02	108	169.4	12,049
9	2026-01-09	163	198.4	13,555
10	2026-01-16	193	227.3	15,061
11	2026-01-23	219	256.3	16,567
12	2026-01-30	205	285.2	18,073
13	2026-02-06	285	314.2	19,579
14	2026-02-13	371	343.2	21,086
15	2026-02-20	384	372.1	22,592
16	2026-02-27	401	401.1	24,098
17	2026-03-06	462	430.1	25,604
18	2026-03-13	483	459.0	27,110
19	2026-03-20	572	488.0	28,616
20	2026-03-27	525	517.0	30,122
21	2026-04-03	515	545.9	31,628
22	2026-04-10	548	574.9	33,134
23	2026-04-17	580	603.8	34,641
24	2026-04-24	616	632.8	36,147
25	2026-05-01	697	661.8	37,653
26	2026-05-08	706	690.7	39,159
27	2026-05-15	677	719.7	40,665
28	2026-05-22	806	748.7	42,171
29	2026-05-29	703	777.6	43,677
30	2026-06-05	840	806.6	45,183
31	2026-06-12	859	835.6	46,690
32	2026-06-19	NA	864.5	48,196
33	2026-06-26	NA	893.5	49,702
34	2026-07-03	NA	922.4	51,208
35	2026-07-10	NA	951.4	52,714

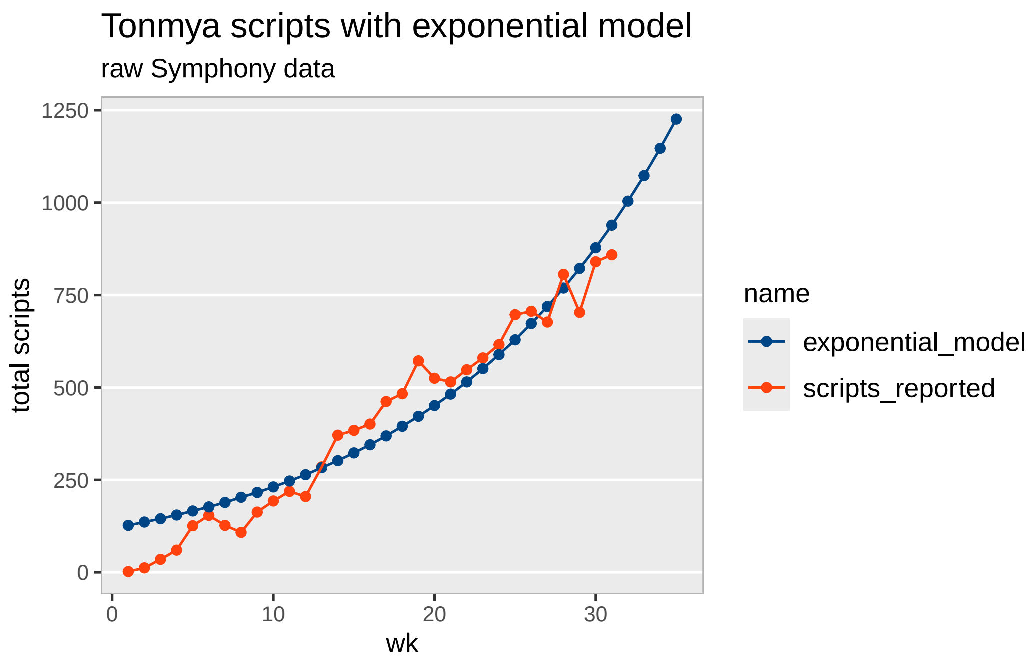


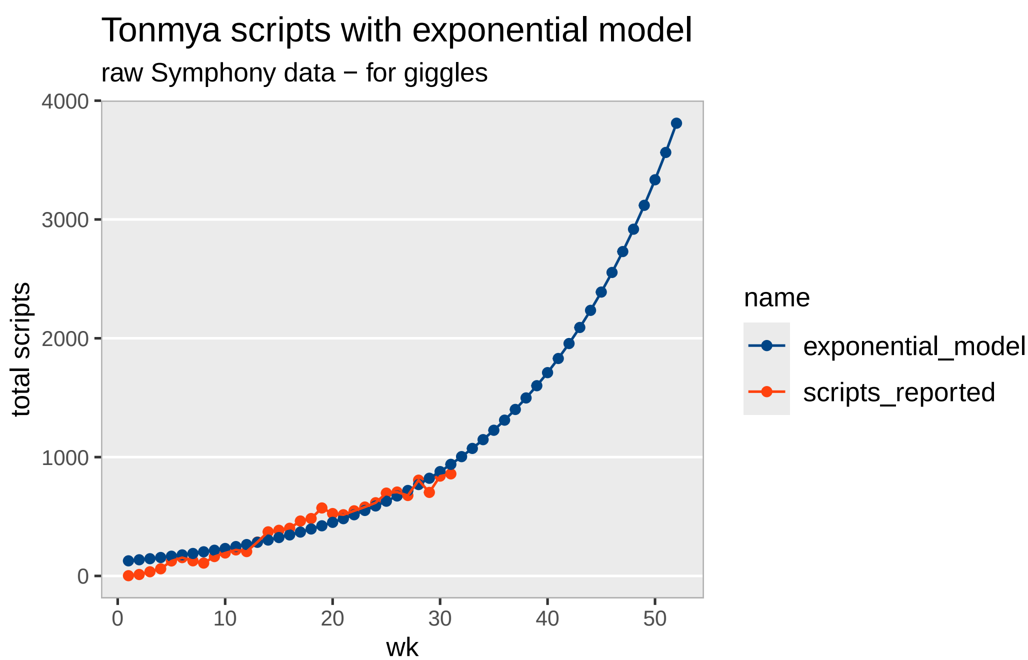
Scripts Projection - at weekly growth rate - ***exponential model***

```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 1.188e+02  1.362e+01   8.723 1.33e-09 ***
r 6.897e-02  4.843e-03  14.242 1.27e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 75.75 on 29 degrees of freedom

Number of iterations to convergence: 10
Achieved convergence tolerance: 3.237e-06
```

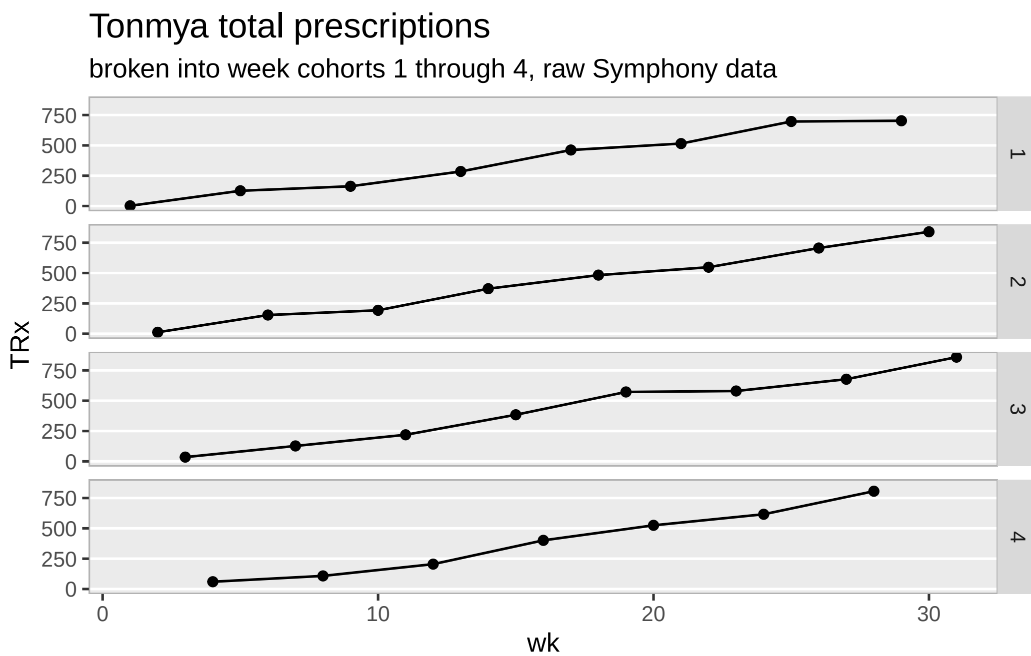
Scripts, Exponential Model

wk	weekly_seq	total reported	exp predicted	annual
1	2025-11-14	2	127	6,602
2	2025-11-21	12	136	7,058
3	2025-11-28	35	145	7,544
4	2025-12-05	60	155	8,065
5	2025-12-12	126	166	8,621
6	2025-12-19	154	177	9,215
7	2025-12-26	127	189	9,851
8	2026-01-02	108	203	10,531
9	2026-01-09	163	216	11,257
10	2026-01-16	193	231	12,033
11	2026-01-23	219	247	12,863
12	2026-01-30	205	264	13,750
13	2026-02-06	285	283	14,699
14	2026-02-13	371	302	15,713
15	2026-02-20	384	323	16,796
16	2026-02-27	401	345	17,955
17	2026-03-06	462	369	19,193
18	2026-03-13	483	395	20,517
19	2026-03-20	572	422	21,932
20	2026-03-27	525	451	23,445
21	2026-04-03	515	482	25,062
22	2026-04-10	548	515	26,790
23	2026-04-17	580	551	28,638
24	2026-04-24	616	589	30,613
25	2026-05-01	697	629	32,725
26	2026-05-08	706	673	34,982
27	2026-05-15	677	719	37,395
28	2026-05-22	806	769	39,974
29	2026-05-29	703	822	42,731
30	2026-06-05	840	878	45,678
31	2026-06-12	859	939	48,829
32	2026-06-19	NA	1004	52,196
33	2026-06-26	NA	1073	55,796
34	2026-07-03	NA	1147	59,645
35	2026-07-10	NA	1226	63,759





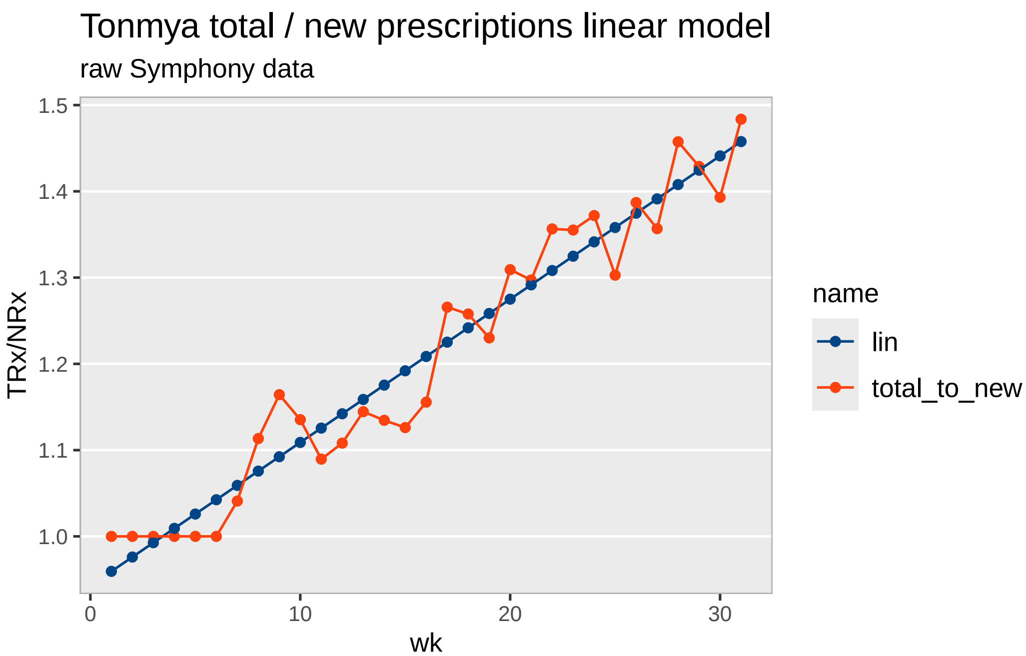
TRx Total Prescriptions by Weekly Cohorts



total / new scripts at weekly rate - *linear model*

```
Call:
lm(formula = total_to_new ~ wk, data = trx_nrx_df)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.065862 -0.034289  0.005617  0.030406  0.071975 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 0.9428349  0.0139840   67.42   <2e-16 ***
wk          0.0166084  0.0007629   21.77   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03799 on 29 degrees of freedom
Multiple R-squared:  0.9423,	Adjusted R-squared:  0.9404 
F-statistic:   474 on 1 and 29 DF,  p-value: < 2.2e-16
```

wk	scripts_reported
1	
1	2
5	126
9	163
13	285
17	462
21	515
25	697
29	703
2	
2	12
6	154
10	193
14	371
18	483
22	548
26	706
30	840
3	
3	35
7	127
11	219
15	384
19	572
23	580
27	677
31	859
4	
4	60
8	108
12	205
16	401
20	525
24	616
28	806

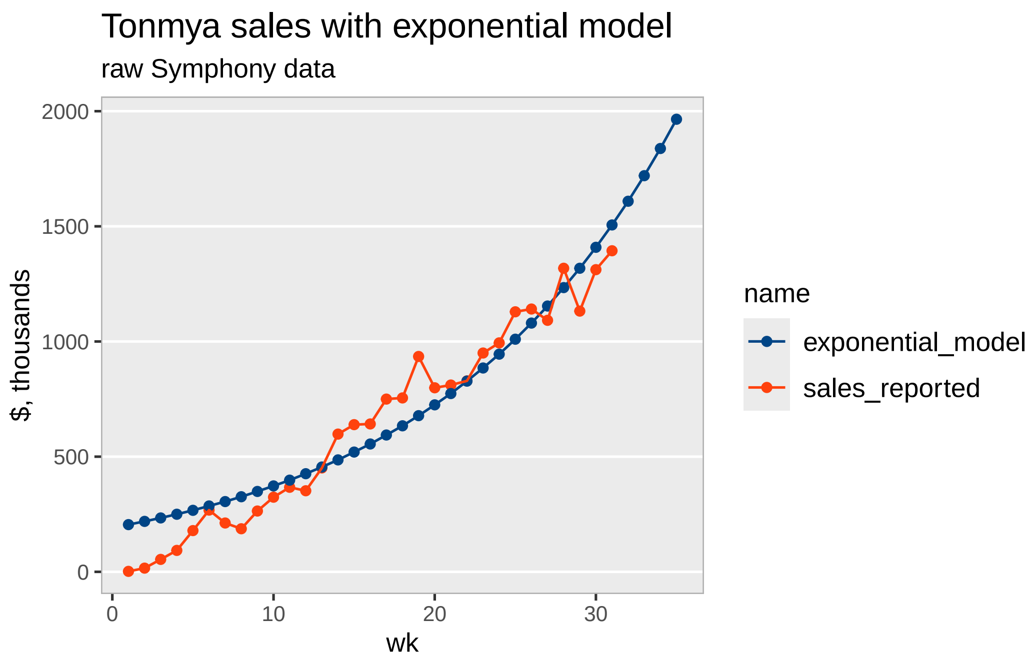


Sales Projection - at weekly growth rate - *exponential model*

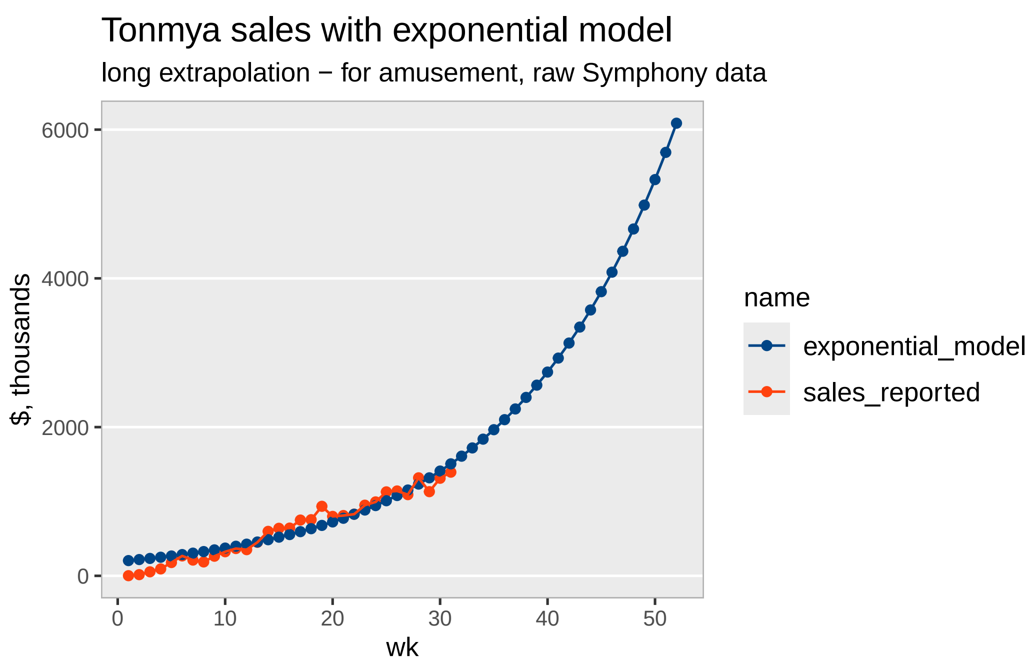
```
Formula: sales ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 1.916e+02  2.194e+01   8.734 1.30e-09 ***
r 6.877e-02  4.838e-03  14.212 1.34e-14 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 121.7 on 29 degrees of freedom

Number of iterations to convergence: 13
Achieved convergence tolerance: 6.542e-06
```

Exponential Model

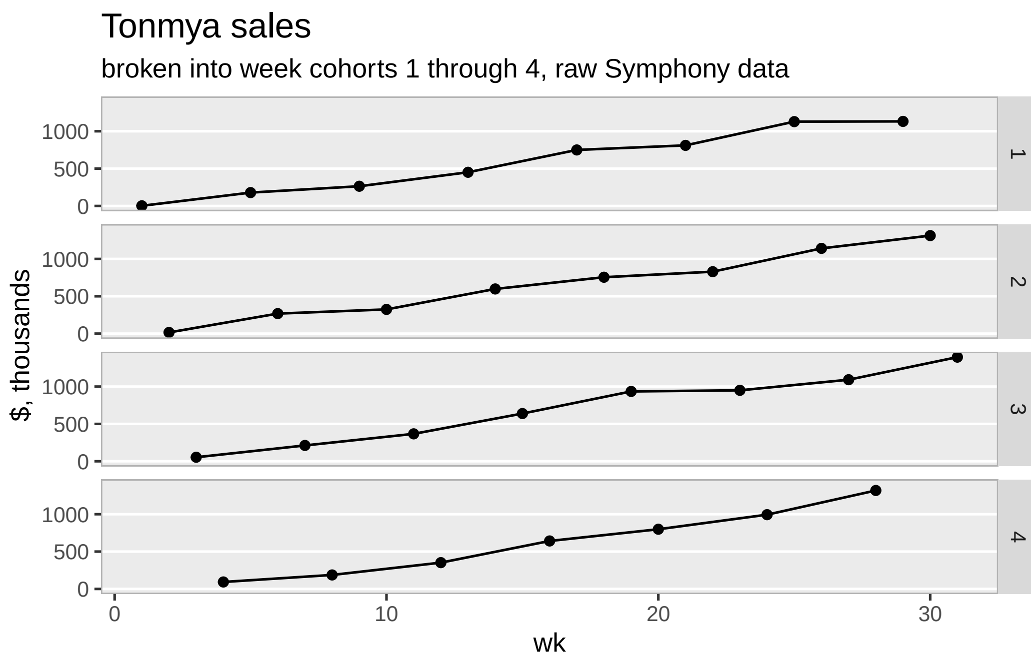
wk	weekly_seq	sales, thousands	exp, thousands	annual, M
1	2025-11-14	$2	$205	$10.6
2	2025-11-21	$16	$219	$11.4
3	2025-11-28	$54	$234	$12.2
4	2025-12-05	$93	$250	$13.0
5	2025-12-12	$179	$267	$13.9
6	2025-12-19	$268	$286	$14.9
7	2025-12-26	$212	$305	$15.9
8	2026-01-02	$187	$326	$17.0
9	2026-01-09	$264	$349	$18.1
10	2026-01-16	$324	$373	$19.4
11	2026-01-23	$367	$398	$20.7
12	2026-01-30	$352	$426	$22.1
13	2026-02-06	$451	$455	$23.7
14	2026-02-13	$598	$486	$25.3
15	2026-02-20	$639	$520	$27.0
16	2026-02-27	$642	$555	$28.9
17	2026-03-06	$750	$594	$30.9
18	2026-03-13	$755	$634	$33.0
19	2026-03-20	$935	$678	$35.3
20	2026-03-27	$799	$725	$37.7
21	2026-04-03	$811	$774	$40.3
22	2026-04-10	$829	$828	$43.0
23	2026-04-17	$950	$885	$46.0
24	2026-04-24	$994	$945	$49.2
25	2026-05-01	$1,129	$1,010	$52.5
26	2026-05-08	$1,141	$1,080	$56.2
27	2026-05-15	$1,092	$1,154	$60.0
28	2026-05-22	$1,318	$1,234	$64.1
29	2026-05-29	$1,132	$1,318	$68.6
30	2026-06-05	$1,312	$1,409	$73.3
31	2026-06-12	$1,394	$1,506	$78.3
32	2026-06-19	NA	$1,609	$83.7
33	2026-06-26	NA	$1,720	$89.4
34	2026-07-03	NA	$1,838	$95.6
35	2026-07-10	NA	$1,965	$102.2



Tonmya sales with exponential model
raw Symphony data
2000
1500
1000
500
0
$, thousands
0
10
20
30
wk
name
exponential_model
sales_reported



Tonmya sales with exponential model
long extrapolation − for amusement, raw Symphony data
6000
4000
2000
0
$, thousands
0
10
20
30
40
50
wk
name
exponential_model
sales_reported

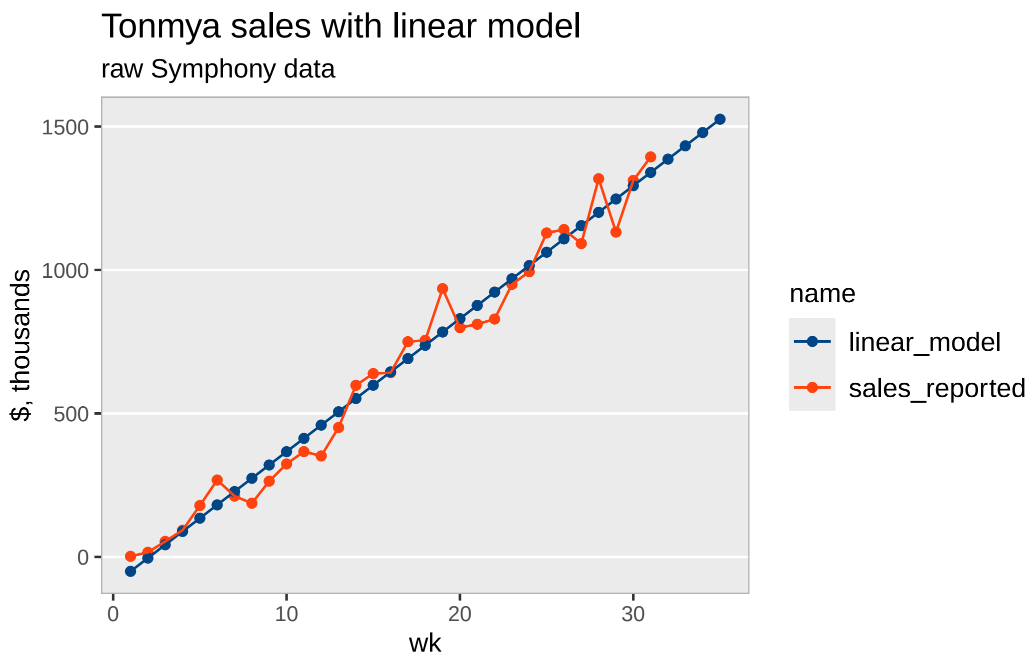


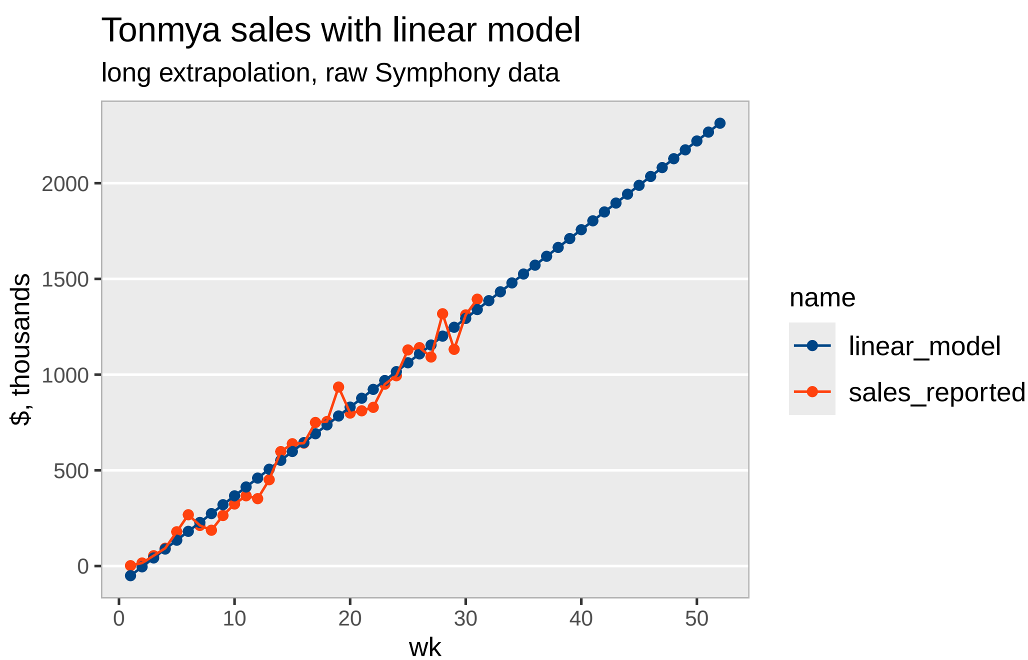
Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
    Min      1Q  Median      3Q     Max
-115.30  -50.42    4.34   44.94  151.16

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   -96.72      24.38  -3.967 0.000438 ***
wk             46.35       1.33  34.840  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 66.25 on 29 degrees of freedom
Multiple R-squared:  0.9767,    Adjusted R-squared:  0.9759
F-statistic:  1214 on 1 and 29 DF,  p-value: < 2.2e-16
```

Linear Model

wk	weekly_seq	sales, thousands	linear, thousands	annual, M
1	2025-11-14	$2	-$50	$2.4
2	2025-11-21	$16	-$4	$4.8
3	2025-11-28	$54	$42	$7.2
4	2025-12-05	$93	$89	$9.6
5	2025-12-12	$179	$135	$12.0
6	2025-12-19	$268	$181	$14.5
7	2025-12-26	$212	$228	$16.9
8	2026-01-02	$187	$274	$19.3
9	2026-01-09	$264	$320	$21.7
10	2026-01-16	$324	$367	$24.1
11	2026-01-23	$367	$413	$26.5
12	2026-01-30	$352	$459	$28.9
13	2026-02-06	$451	$506	$31.3
14	2026-02-13	$598	$552	$33.7
15	2026-02-20	$639	$598	$36.1
16	2026-02-27	$642	$645	$38.6
17	2026-03-06	$750	$691	$41.0
18	2026-03-13	$755	$737	$43.4
19	2026-03-20	$935	$784	$45.8
20	2026-03-27	$799	$830	$48.2
21	2026-04-03	$811	$877	$50.6
22	2026-04-10	$829	$923	$53.0
23	2026-04-17	$950	$969	$55.4
24	2026-04-24	$994	$1,016	$57.8
25	2026-05-01	$1,129	$1,062	$60.2
26	2026-05-08	$1,141	$1,108	$62.7
27	2026-05-15	$1,092	$1,155	$65.1
28	2026-05-22	$1,318	$1,201	$67.5
29	2026-05-29	$1,132	$1,247	$69.9
30	2026-06-05	$1,312	$1,294	$72.3
31	2026-06-12	$1,394	$1,340	$74.7
32	2026-06-19	NA	$1,386	$77.1
33	2026-06-26	NA	$1,433	$79.5
34	2026-07-03	NA	$1,479	$81.9
35	2026-07-10	NA	$1,525	$84.3





Refills - at weekly rate - *linear model*

```
Call:
lm(formula = refills ~ wk, data = trx_nrx_df)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-38.223 -12.911  -2.797  12.679  52.219 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -56.7290     9.2321  -6.145 1.07e-06 ***
wk            9.1968     0.5037  18.260  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 25.08 on 29 degrees of freedom
Multiple R-squared:   0.92, Adjusted R-squared:  0.9172 
F-statistic: 333.4 on 1 and 29 DF,  p-value: < 2.2e-16
```